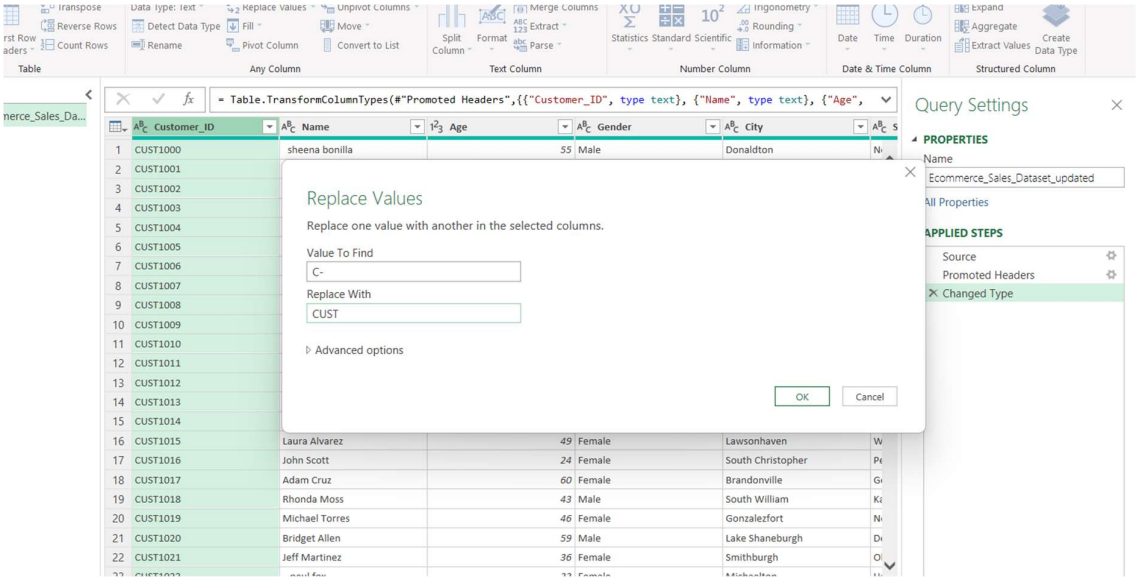
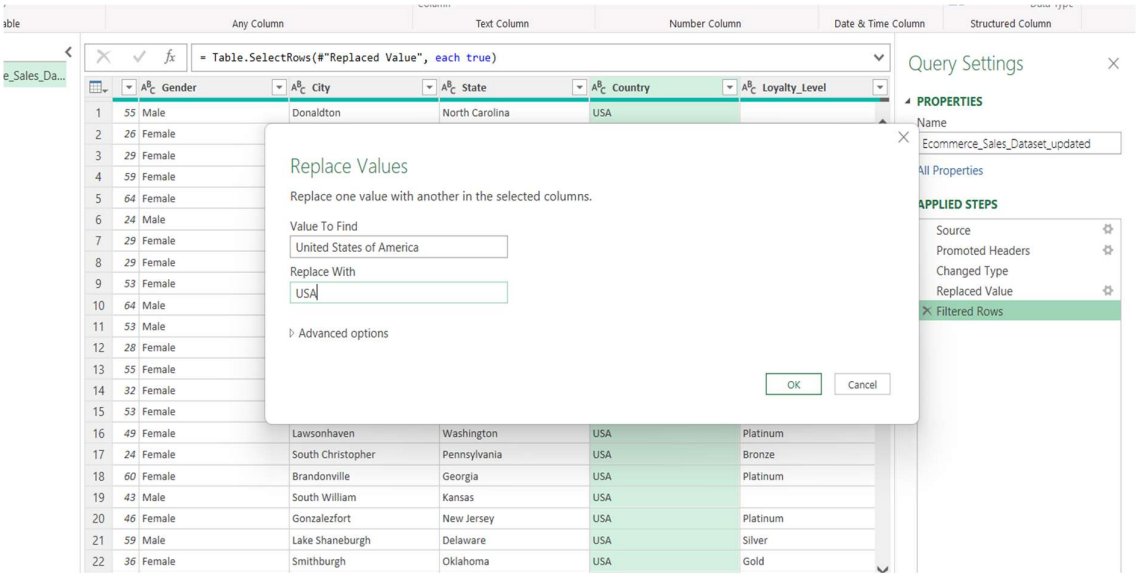
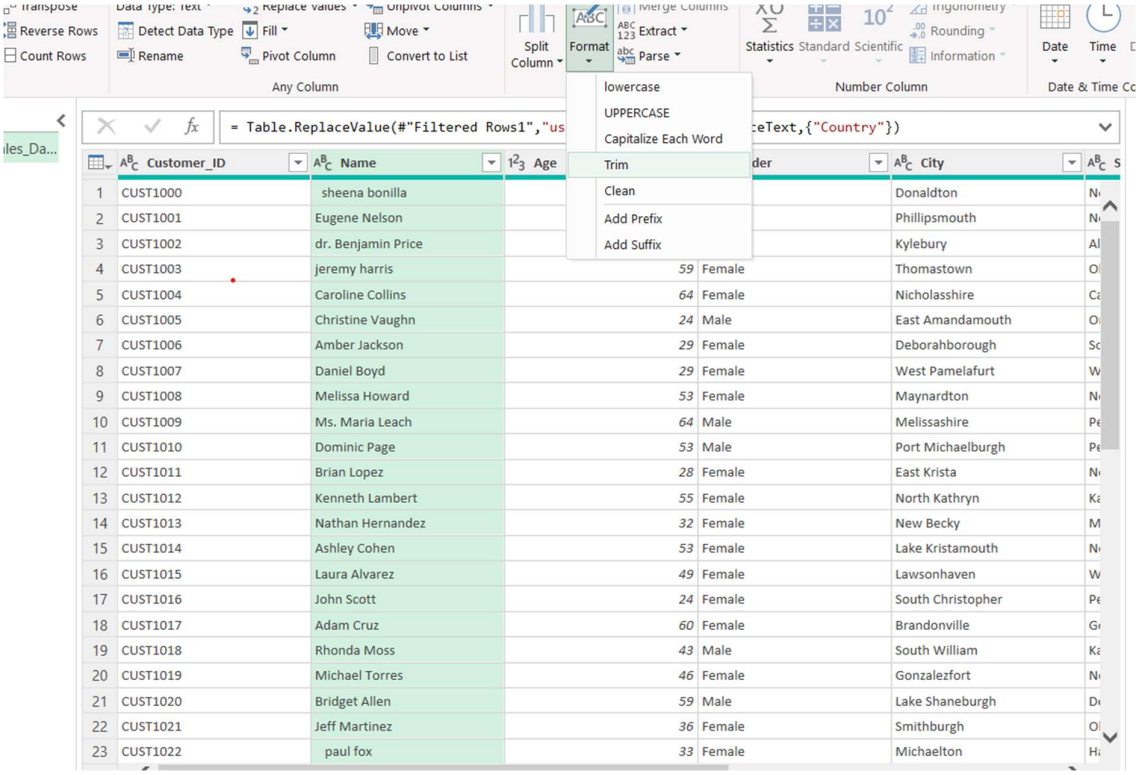
1) Observations and screenshots for Customer_Dim dataset

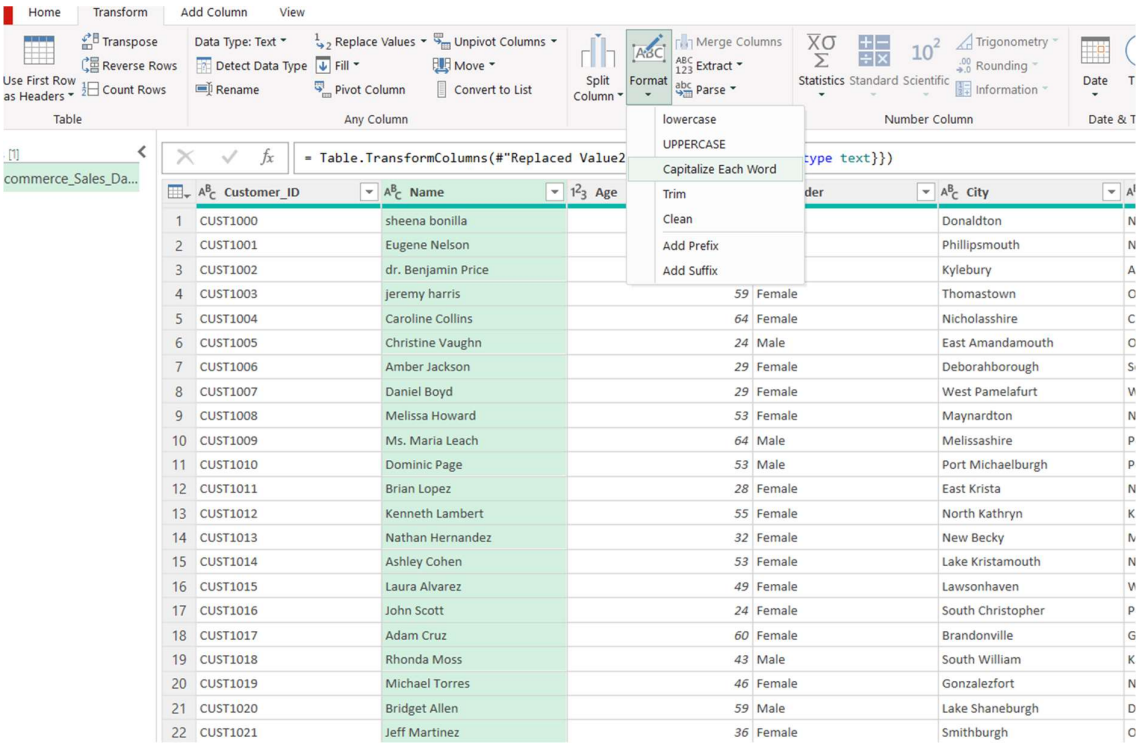
Step 1 : Replaced the word C- into CUST in customer ID column and replaced the United states of America in to USA in country column using find and replace



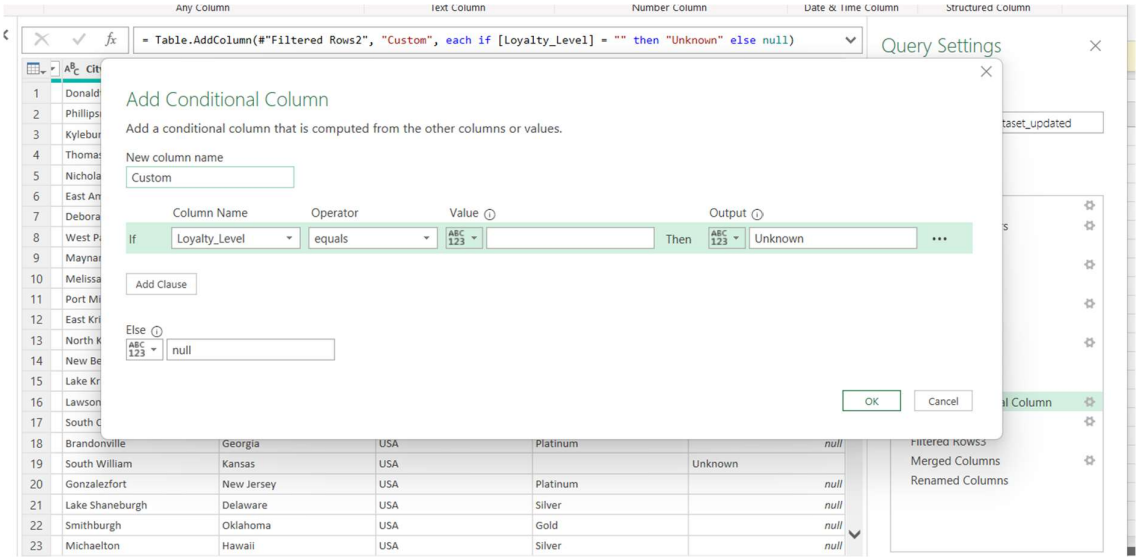


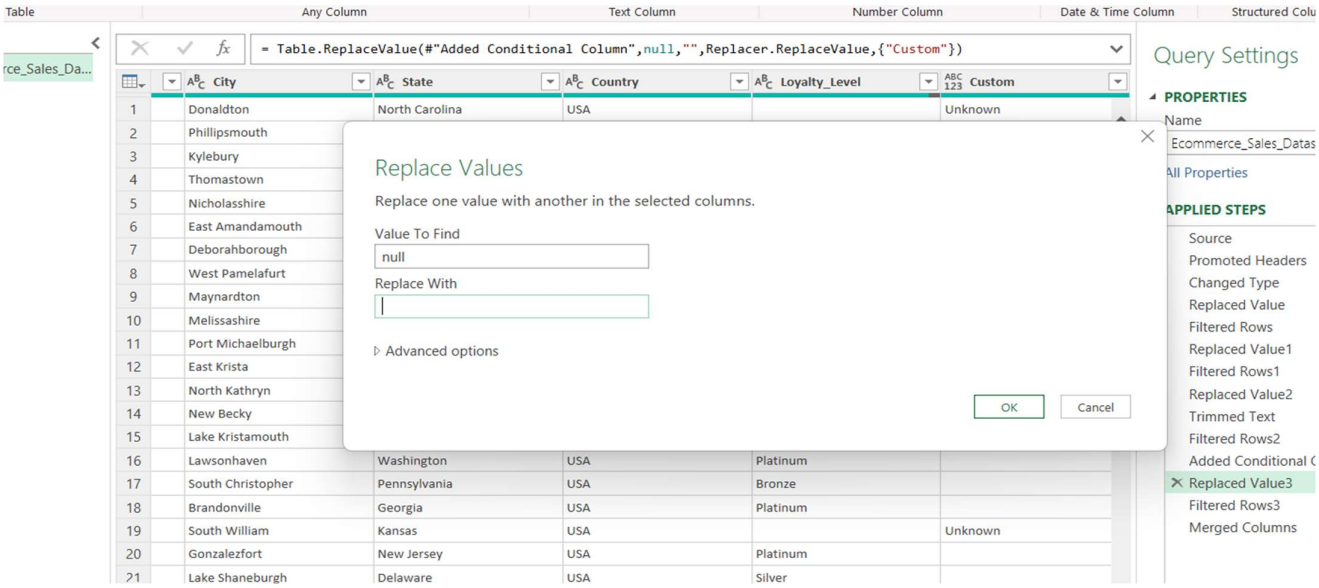
Step 2 : Used Trim function to adjust the spaces in name column for some name and capitalized each word using capitalized each word in text format tab

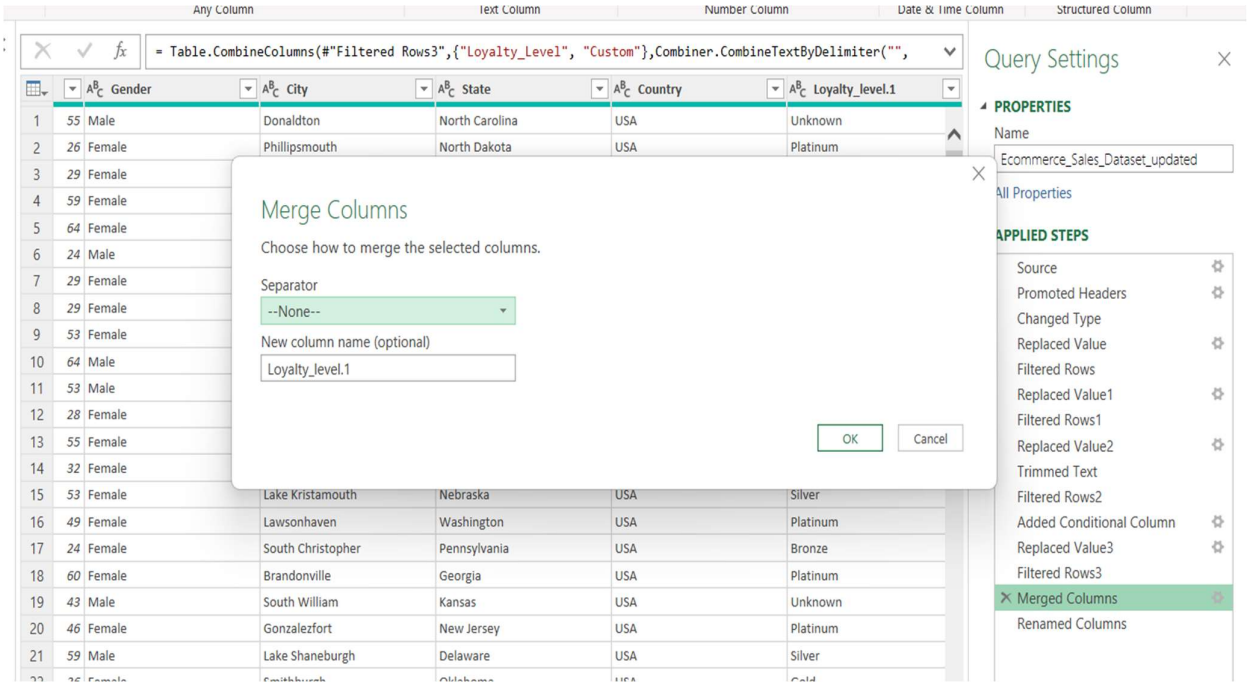




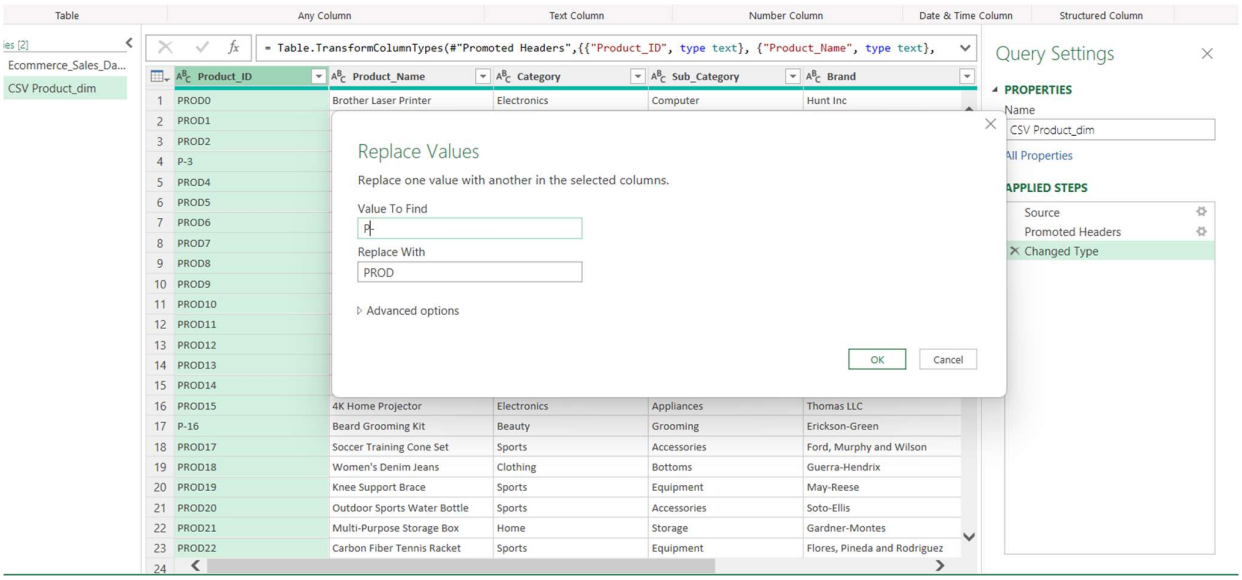
Step 3 – 1) Replaced blank value with unknown in loyalty_level column by first using conditional formatting and provided the below output – 2) Then replaced the null word with blank using find and replace option 3) Then merged the columns using merge columns option



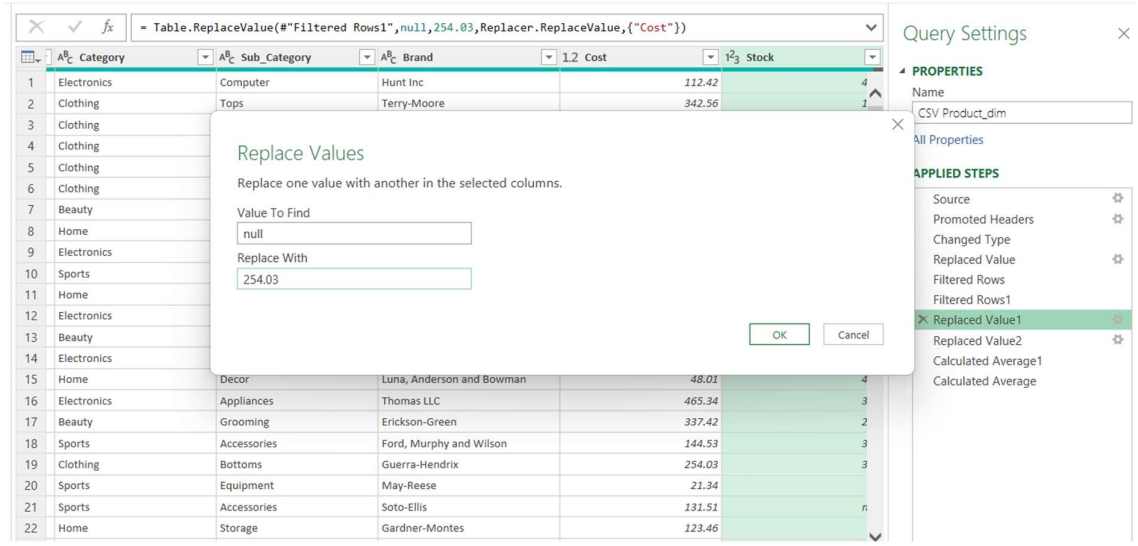


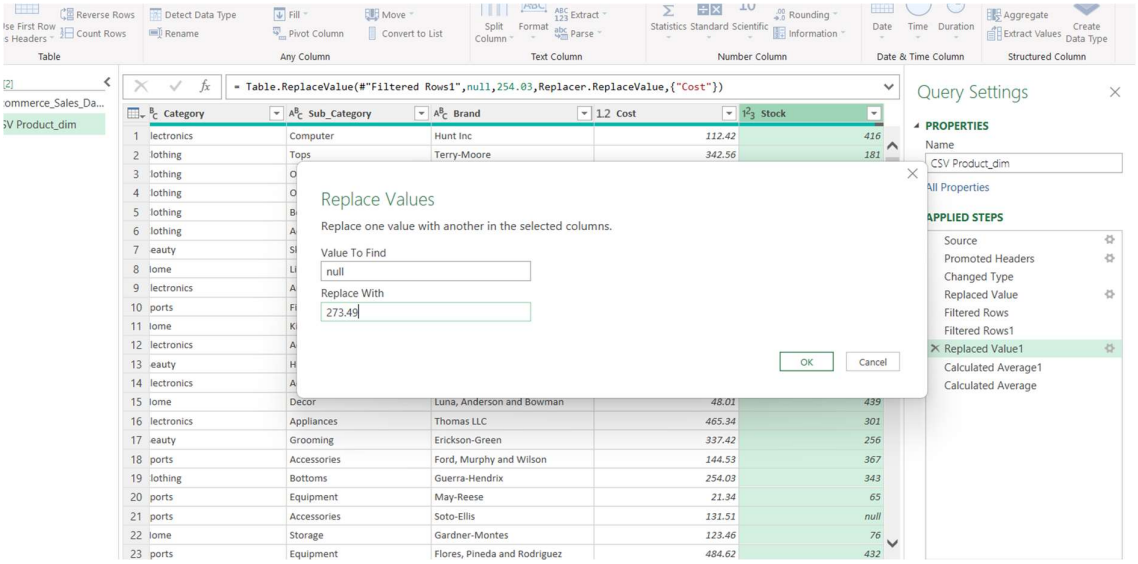


2) Observations and screenshots for Product_Dim dataset

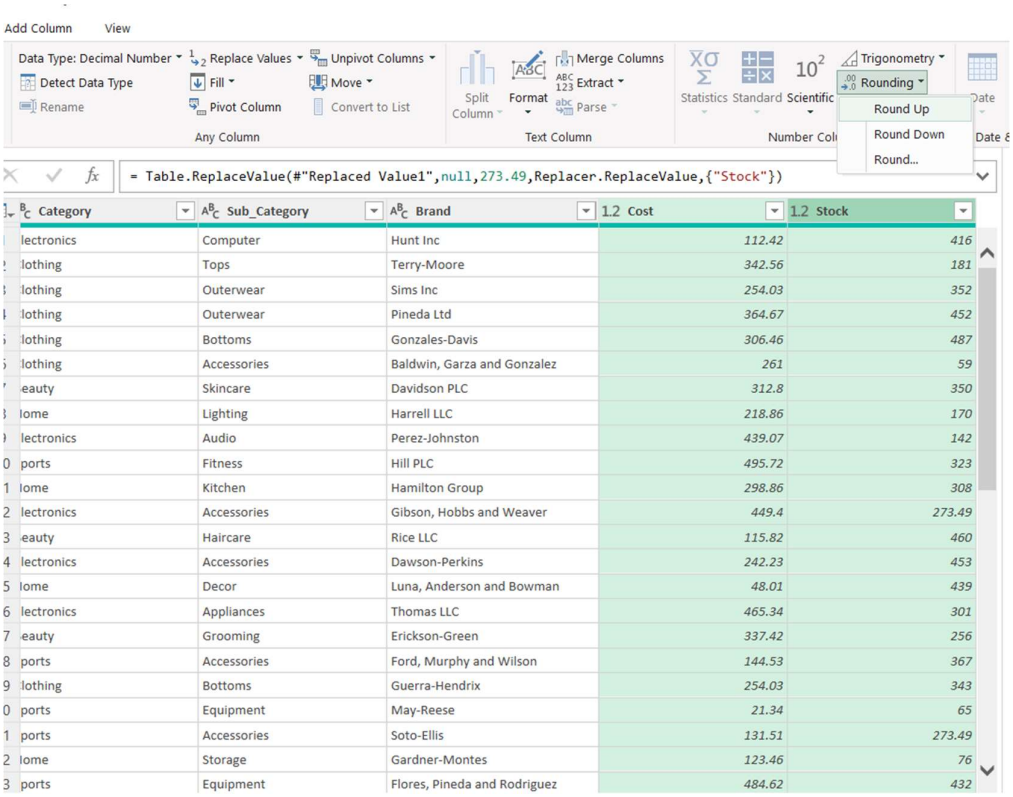


To find the average for cost and stock column





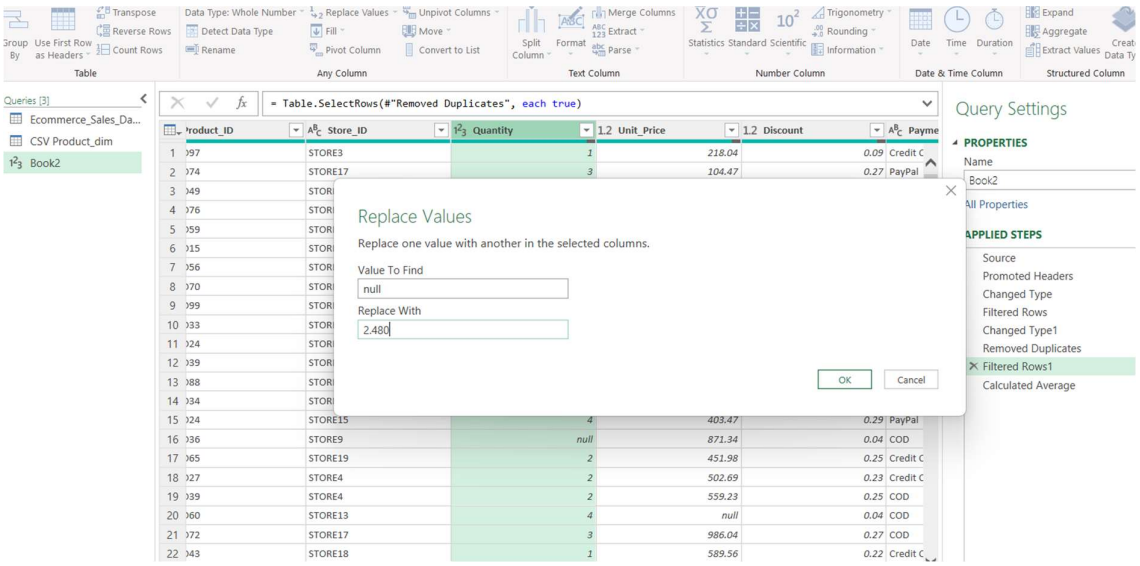
Cost and stock value has been rounded up using rounding function



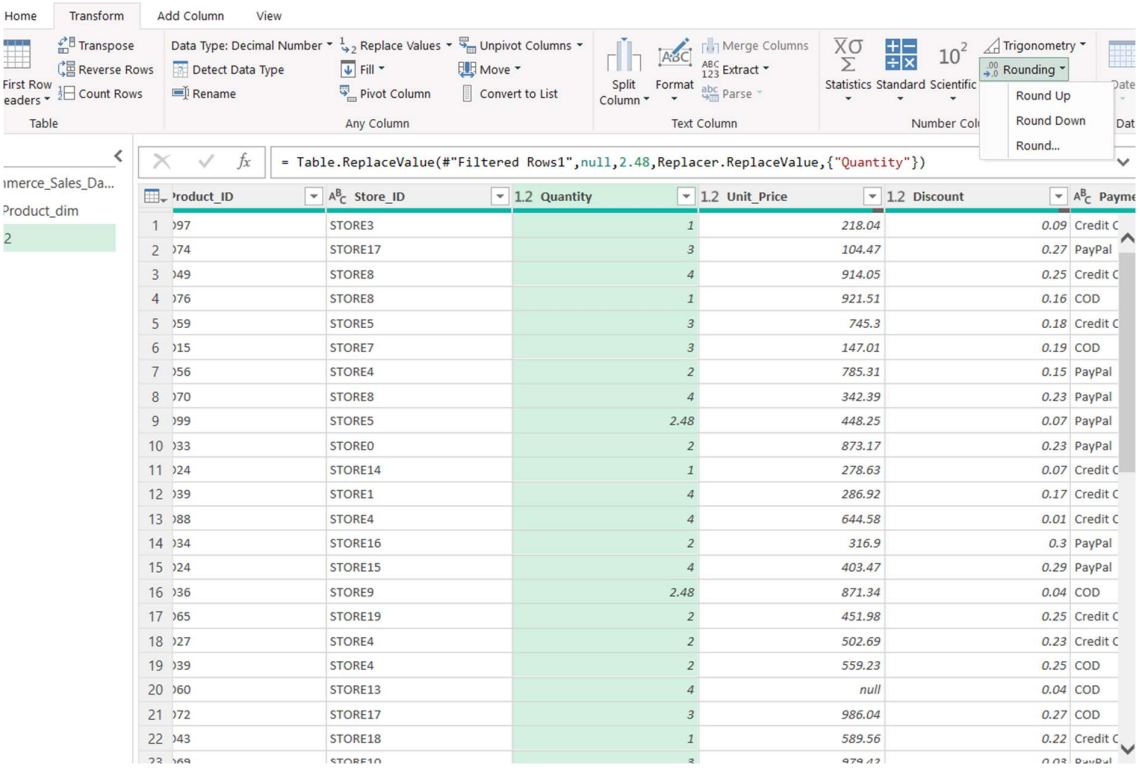
	Category	Sub_Category	Brand	Cost	Stock
	lectronics	Computer	Hunt Inc	112.42	416
	lothing	Tops	Terry-Moore	342.56	181
	lothing	Outerwear	Sims Inc	254.03	352
	lothing	Outerwear	Pineda Ltd	364.67	452
	lothing	Bottoms	Gonzales-Davis	306.46	487
	lothing	Accessories	Baldwin, Garza and Gonzalez	261	59
	eauty	Skincare	Davidson PLC	312.8	350
	lome	Lighting	Harrell LLC	218.86	170
	lectronics	Audio	Perez-Johnston	439.07	142
0	ports	Fitness	Hill PLC	495.72	323
1	lome	Kitchen	Hamilton Group	298.86	308
2	lectronics	Accessories	Gibson, Hobbs and Weaver	449.4	273.49
3	eauty	Haircare	Rice LLC	115.82	460
4	lectronics	Accessories	Dawson-Perkins	242.23	453
5	lome	Decor	Luna, Anderson and Bowman	48.01	439
6	lectronics	Appliances	Thomas LLC	465.34	301
7	eauty	Grooming	Erickson-Green	337.42	256
8	ports	Accessories	Ford, Murphy and Wilson	144.53	367
9	lothing	Bottoms	Guerra-Hendrix	254.03	343
0	ports	Equipment	May-Reese	21.34	65
1	ports	Accessories	Soto-Ellis	131.51	273.49
2	lome	Storage	Gardner-Montes	123.46	76
3	ports	Equipment	Flores, Pineda and Rodriguez	484.62	432

Observations and screenshots for Sales_fact

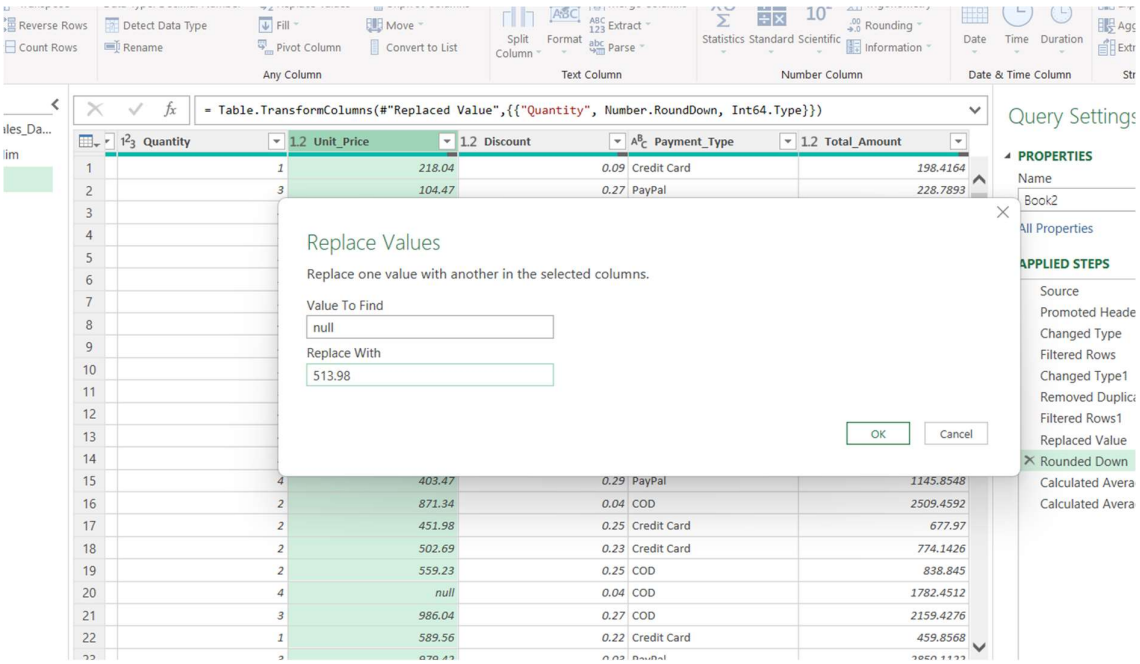
Took average for quantity column and replaced null value with the average value 2.480 and used rounding to round the value to 2



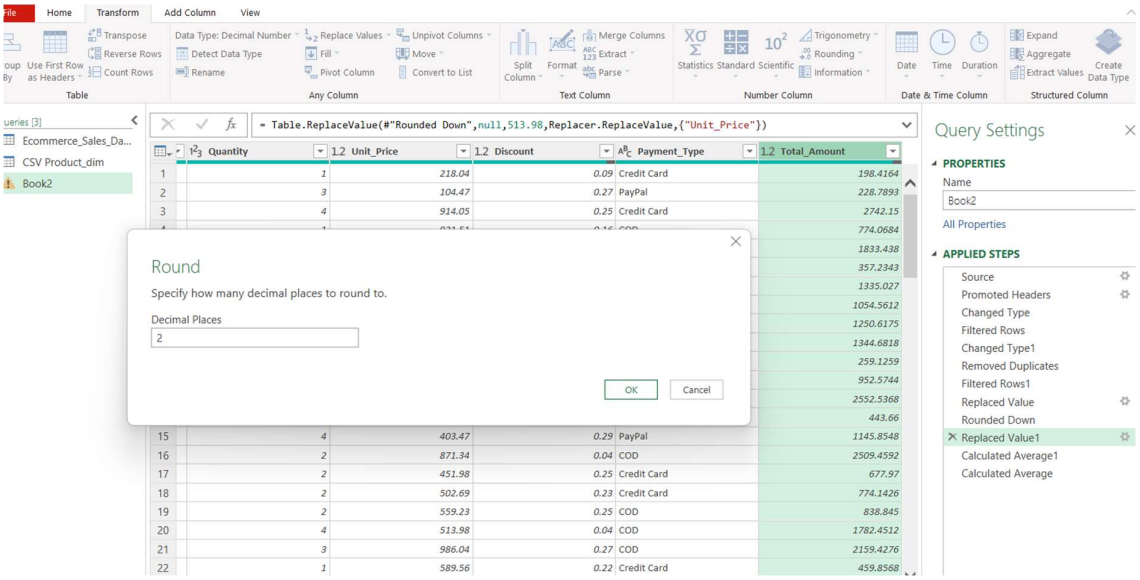
	roduct_ID	Store_ID	Quantity	Unit_Price	Discount	Payme
1	)97	STORE3	1	218.04	0.09	Credit C
2	)74	STORE17	3	104.47	0.27	PayPal
3	)49	STORI				
4	)76	STORI				
5	)59	STORI				
6	)15	STORI				
7	)56	STORI				
8	)70	STORI				
9	)99	STORI				
10	)33	STORI				
11	)24	STORI				
12	)39	STORI				
13	)88	STORI				
14	)34	STORI				
15	)24	STORE15	4	403.47	0.29	PayPal
16	)36	STORE9	null	871.34	0.04	COD
17	)65	STORE19	2	451.98	0.25	Credit C
18	)27	STORE4	2	502.69	0.23	Credit C
19	)39	STORE4	2	559.23	0.25	COD
20	)60	STORE13	4	null	0.04	COD
21	)72	STORE17	3	986.04	0.27	COD
22	)43	STORE18	1	589.56	0.22	Credit C

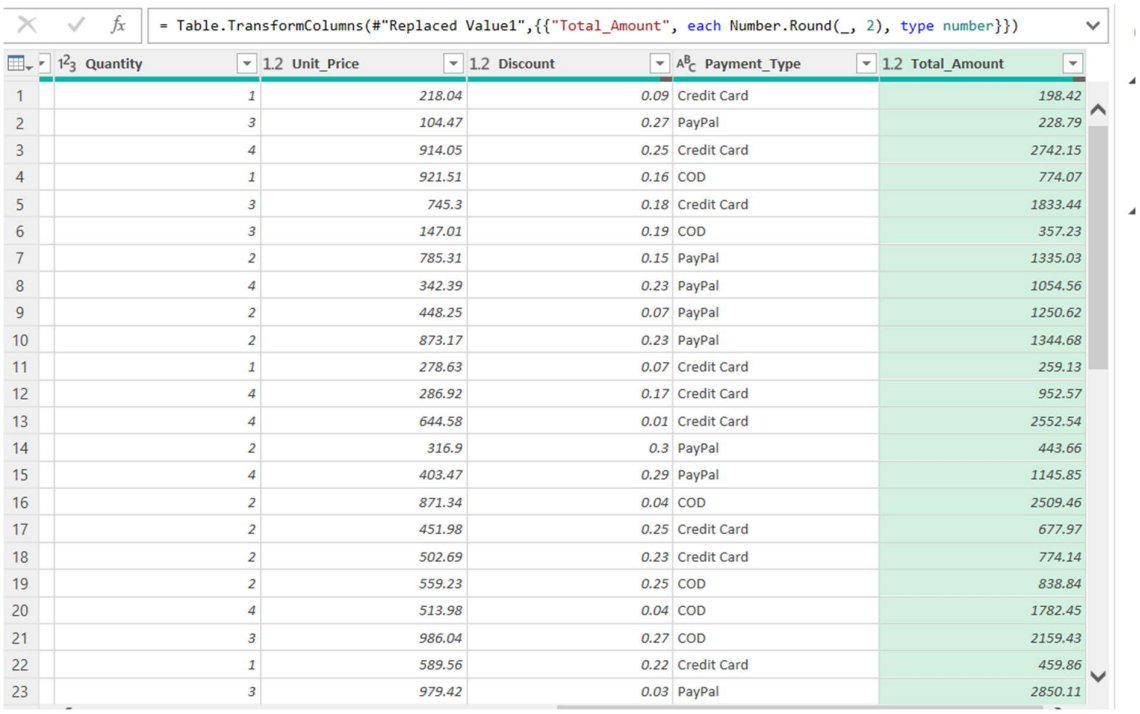


Took average for unit price column and replaced null value with the average value 513.98



Using rounding option, we rounded the total amount column by using the decimal places as 2

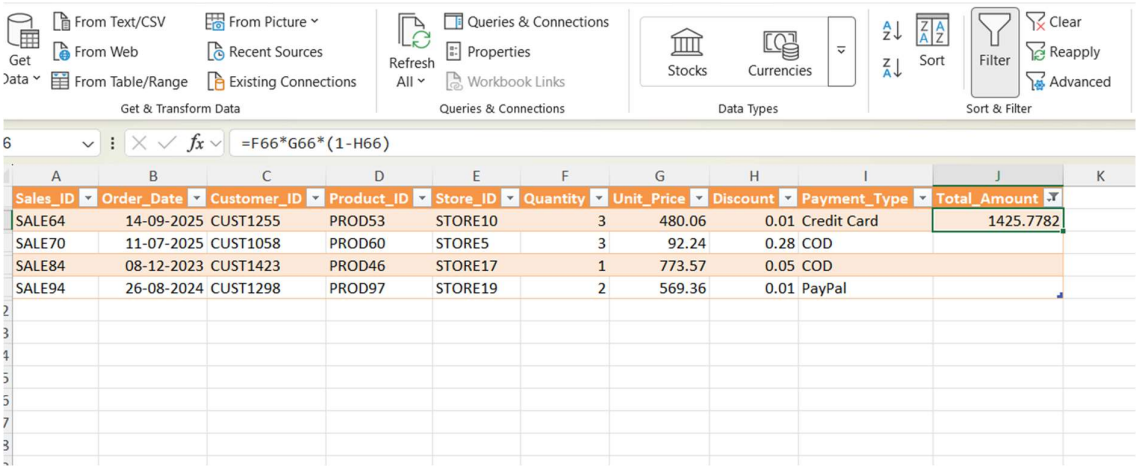




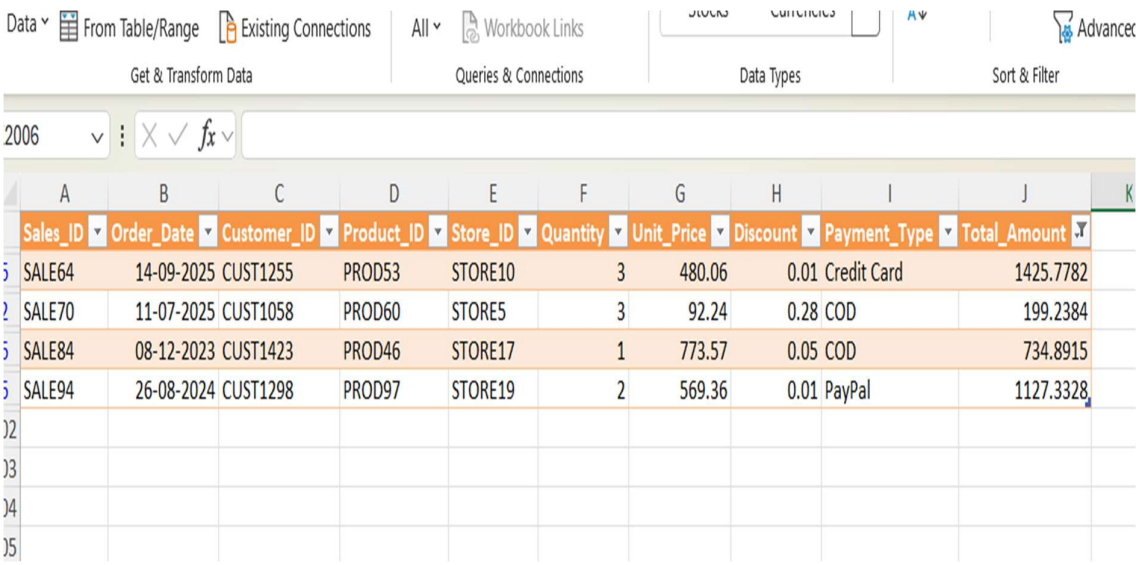
= Table.TransformColumns(#"Replaced Value1",{{"Total_Amount", each Number.Round(_, 2), type number}})

	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
1	1	218.04	0.09	Credit Card	198.42
2	3	104.47	0.27	PayPal	228.79
3	4	914.05	0.25	Credit Card	2742.15
4	1	921.51	0.16	COD	774.07
5	3	745.3	0.18	Credit Card	1833.44
6	3	147.01	0.19	COD	357.23
7	2	785.31	0.15	PayPal	1335.03
8	4	342.39	0.23	PayPal	1054.56
9	2	448.25	0.07	PayPal	1250.62
10	2	873.17	0.23	PayPal	1344.68
11	1	278.63	0.07	Credit Card	259.13
12	4	286.92	0.17	Credit Card	952.57
13	4	644.58	0.01	Credit Card	2552.54
14	2	316.9	0.3	PayPal	443.66
15	4	403.47	0.29	PayPal	1145.85
16	2	871.34	0.04	COD	2509.46
17	2	451.98	0.25	Credit Card	677.97
18	2	502.69	0.23	Credit Card	774.14
19	2	559.23	0.25	COD	838.84
20	4	513.98	0.04	COD	1782.45
21	3	986.04	0.27	COD	2159.43
22	1	589.56	0.22	Credit Card	459.86
23	3	979.42	0.03	PayPal	2850.11

Found that there are 4 blank value in total and we used the formula =F66*G66*(1-H66) to find the total amount

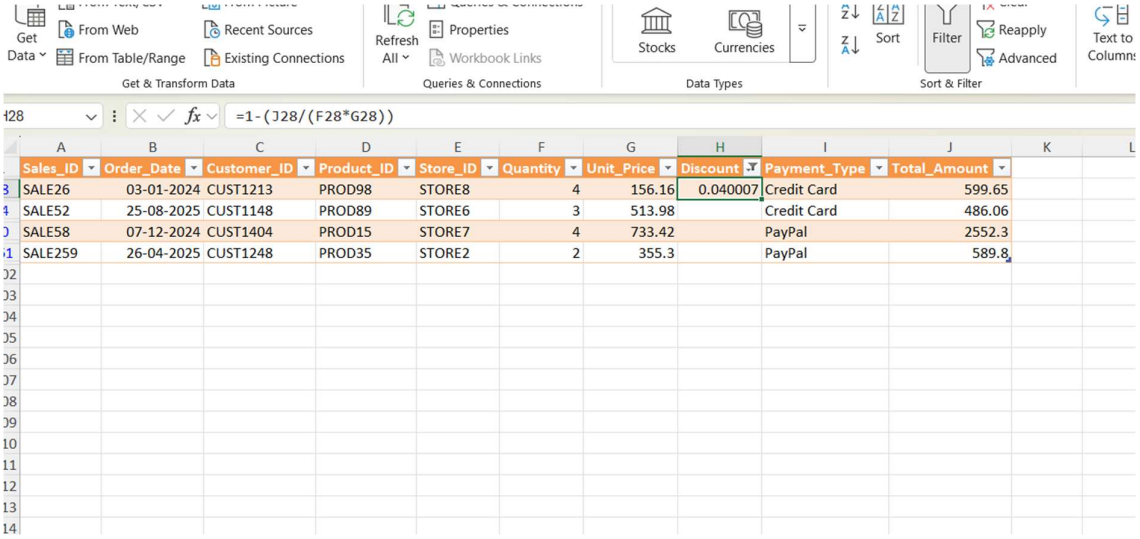


Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE64	14-09-2025	CUST1255	PROD53	STORE10	3	480.06	0.01	Credit Card	1425.7782
SALE70	11-07-2025	CUST1058	PROD60	STORE5	3	92.24	0.28	COD	
SALE84	08-12-2023	CUST1423	PROD46	STORE17	1	773.57	0.05	COD	
SALE94	26-08-2024	CUST1298	PROD97	STORE19	2	569.36	0.01	PayPal	

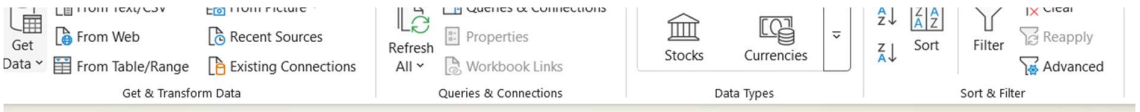


Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE64	14-09-2025	CUST1255	PROD53	STORE10	3	480.06	0.01	Credit Card	1425.7782
SALE70	11-07-2025	CUST1058	PROD60	STORE5	3	92.24	0.28	COD	199.2384
SALE84	08-12-2023	CUST1423	PROD46	STORE17	1	773.57	0.05	COD	734.8915
SALE94	26-08-2024	CUST1298	PROD97	STORE19	2	569.36	0.01	PayPal	1127.3328

Found that there are 4 blank value in total for discount column and we used the formula =1-(J28/(F28*G28)) to find the discount value

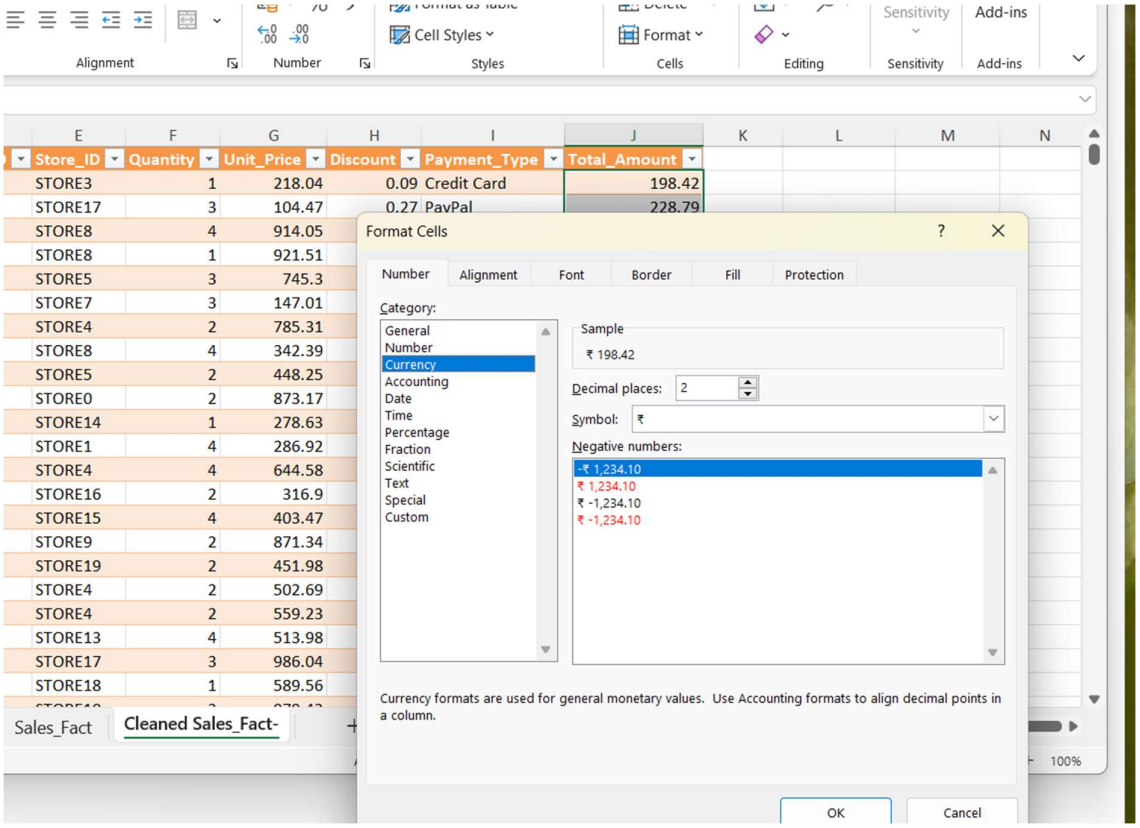


	A	B	C	D	E	F	G	H	I	J	K
	Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	
3	SALE26	03-01-2024	CUST1213	PROD98	STORE8	4	156.16	0.040007	Credit Card	599.65	
4	SALE52	25-08-2025	CUST1148	PROD89	STORE6	3	513.98		Credit Card	486.06	
0	SALE58	07-12-2024	CUST1404	PROD15	STORE7	4	733.42		PayPal	2552.3	
1	SALE259	26-04-2025	CUST1248	PROD35	STORE2	2	355.3		PayPal	589.8	
02											
03											
04											
05											
06											
07											
08											
09											
10											
11											
12											
13											
14											



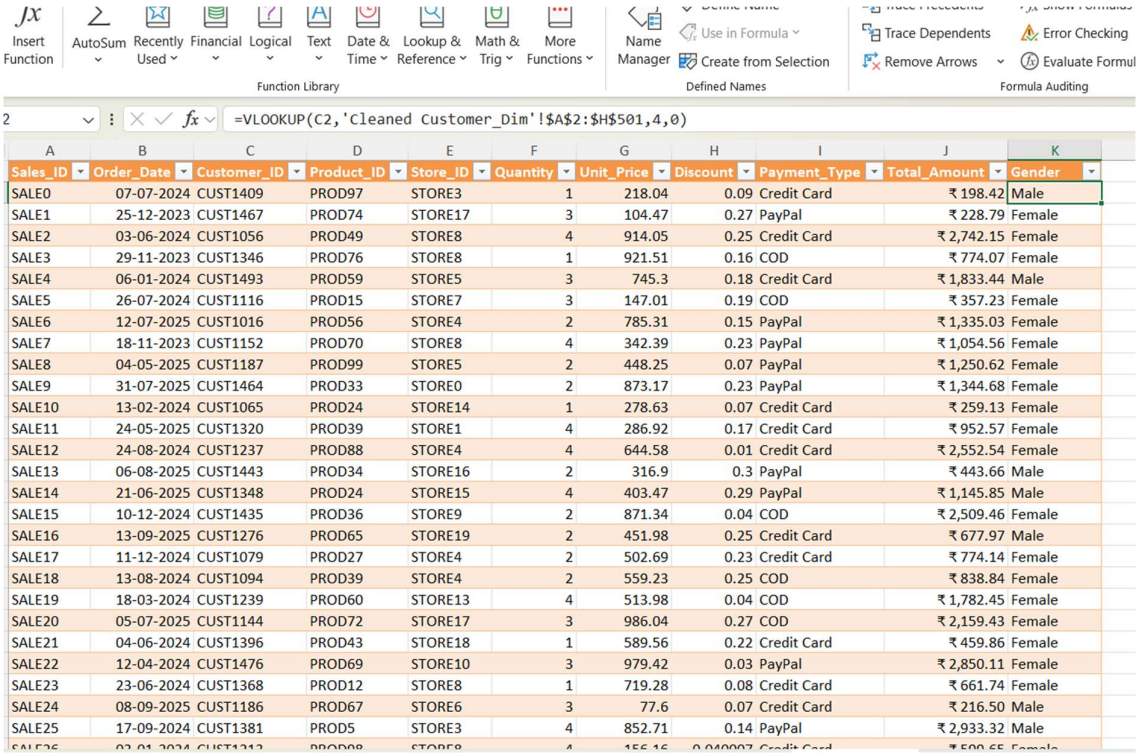
	A	B	C	D	E	F	G	H	I	J	K
	Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	
	SALE26	03-01-2024	CUST1213	PROD98	STORE8	4	156.16	0.040007	Credit Card	599.65	
	SALE52	25-08-2025	CUST1148	PROD89	STORE6	3	513.98	0.6847737	Credit Card	486.06	
	SALE58	07-12-2024	CUST1404	PROD15	STORE7	4	733.42	0.1300005	PayPal	2552.3	
1	SALE259	26-04-2025	CUST1248	PROD35	STORE2	2	355.3	0.1699972	PayPal	589.8	
2											
3											
4											
5											
6											
7											
8											
9											
0											
1											

Converted total amount column into currency format



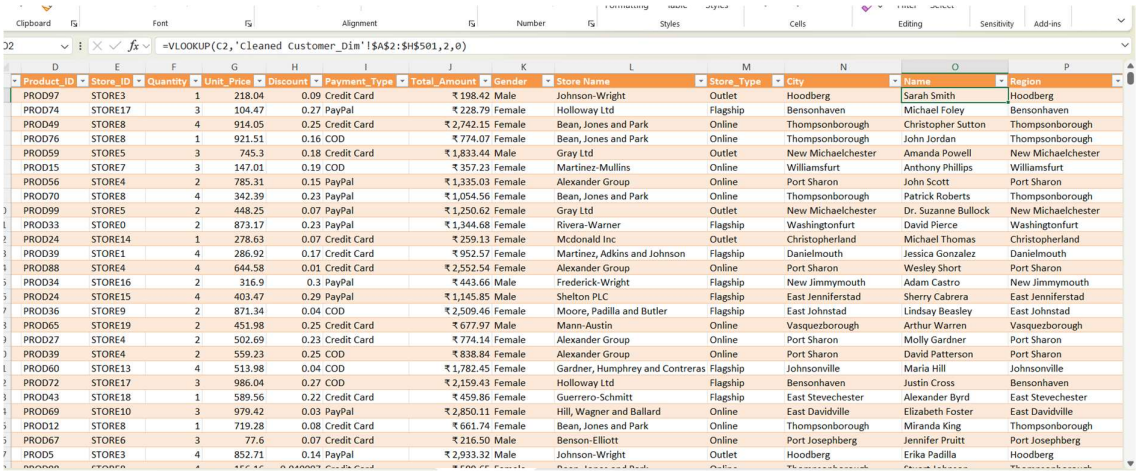
E	F	G	H	I	J	K	L	M	N
tore_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount				
TORE3	1	218.04	0.09	Credit Card	₹ 198.42				
TORE17	3	104.47	0.27	PayPal	₹ 228.79				
TORE8	4	914.05	0.25	Credit Card	₹ 2,742.15				
TORE8	1	921.51	0.16	COD	₹ 774.07				
TORE5	3	745.3	0.18	Credit Card	₹ 1,833.44				
TORE7	3	147.01	0.19	COD	₹ 357.23				
TORE4	2	785.31	0.15	PayPal	₹ 1,335.03				
TORE8	4	342.39	0.23	PayPal	₹ 1,054.56				
TORE5	2	448.25	0.07	PayPal	₹ 1,250.62				
TORE0	2	873.17	0.23	PayPal	₹ 1,344.68				
TORE14	1	278.63	0.07	Credit Card	₹ 259.13				
TORE1	4	286.92	0.17	Credit Card	₹ 952.57				
TORE4	4	644.58	0.01	Credit Card	₹ 2,552.54				
TORE16	2	316.9	0.3	PayPal	₹ 443.66				
TORE15	4	403.47	0.29	PayPal	₹ 1,145.85				
TORE9	2	871.34	0.04	COD	₹ 2,509.46				
TORE19	2	451.98	0.25	Credit Card	₹ 677.97				
TORE4	2	502.69	0.23	Credit Card	₹ 774.14				
TORE4	2	559.23	0.25	COD	₹ 838.84				
TORE13	4	513.98	0.04	COD	₹ 1,782.45				
TORE17	3	986.04	0.27	COD	₹ 2,159.43				
TORE18	1	589.56	0.22	Credit Card	₹ 459.86				

By using VLOOKUP I took the customer gender date from cleaned customer dim sheet by applying the formula **=VLOOKUP(C2,'Cleaned Customer_Dim'!$A$2:$H$501,4,0)**



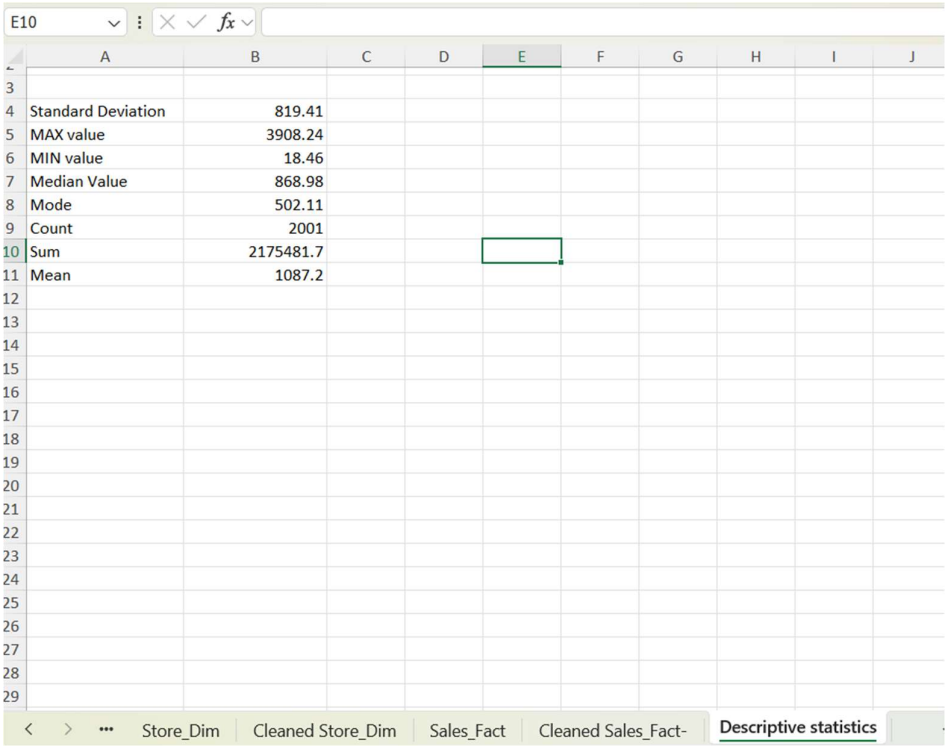
Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Gender
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	₹198.42	Male
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	104.47	0.27	PayPal	₹228.79	Female
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	914.05	0.25	Credit Card	₹2,742.15	Female
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	921.51	0.16	COD	₹774.07	Female
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	745.3	0.18	Credit Card	₹1,833.44	Male
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	147.01	0.19	COD	₹357.23	Female
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	785.31	0.15	PayPal	₹1,335.03	Female
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	342.39	0.23	PayPal	₹1,054.56	Female
SALE8	04-05-2025	CUST1187	PROD99	STORE5	2	448.25	0.07	PayPal	₹1,250.62	Female
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	873.17	0.23	PayPal	₹1,344.68	Female
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	278.63	0.07	Credit Card	₹259.13	Female
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	286.92	0.17	Credit Card	₹952.57	Female
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	644.58	0.01	Credit Card	₹2,552.54	Female
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	316.9	0.3	PayPal	₹443.66	Male
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	₹1,145.85	Male
SALE15	10-12-2024	CUST1435	PROD36	STORE9	2	871.34	0.04	COD	₹2,509.46	Female
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	451.98	0.25	Credit Card	₹677.97	Male
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	502.69	0.23	Credit Card	₹774.14	Female
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	559.23	0.25	COD	₹838.84	Female
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	513.98	0.04	COD	₹1,782.45	Female
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	986.04	0.27	COD	₹2,159.43	Female
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	589.56	0.22	Credit Card	₹459.86	Female
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	979.42	0.03	PayPal	₹2,850.11	Female
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	719.28	0.08	Credit Card	₹661.74	Female
SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	77.6	0.07	Credit Card	₹216.50	Male
SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	852.71	0.14	PayPal	₹2,933.32	Male

Also gathered other details from other cleaned dim sheets using VLOOKUP



Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Gender	Store Name	Store_Type	City	Name	Region
PROD97	STORE3	1	218.04	0.09	Credit Card	₹198.42	Male	Johnson-Wright	Outlet	Hoodberg	Sarah Smith	Hoodberg
PROD74	STORE17	3	104.47	0.27	PayPal	₹228.79	Female	Holloway Ltd	Flagship	Bensonhaven	Michael Foley	Bensonhaven
PROD49	STORE8	4	914.05	0.25	Credit Card	₹2,742.15	Female	Bean, Jones and Park	Online	Thompsonborough	Christopher Sutton	Thompsonborough
PROD76	STORE8	1	921.51	0.16	COD	₹774.07	Female	Bean, Jones and Park	Online	Thompsonborough	John Jordan	Thompsonborough
PROD59	STORE5	3	745.3	0.18	Credit Card	₹1,833.44	Male	Gray Ltd	Outlet	New Michaelchester	Amanda Powell	New Michaelchester
PROD15	STORE7	3	147.01	0.19	COD	₹357.23	Female	Martinez-Mullins	Online	Williamsfurt	Anthony Phillips	Williamsfurt
PROD56	STORE4	2	785.31	0.15	PayPal	₹1,335.03	Female	Alexander Group	Online	Port Sharon	John Scott	Port Sharon
PROD70	STORE8	4	342.39	0.23	PayPal	₹1,054.56	Female	Bean, Jones and Park	Online	Thompsonborough	Patrick Roberts	Thompsonborough
PROD99	STORE5	2	448.25	0.07	PayPal	₹1,250.62	Female	Gray Ltd	Outlet	New Michaelchester	Dr. Suzanne Bullock	New Michaelchester
PROD33	STORE0	2	873.17	0.23	PayPal	₹1,344.68	Female	Rivera-Warner	Flagship	Washingtonfurt	David Pierce	Washingtonfurt
PROD24	STORE14	1	278.63	0.07	Credit Card	₹259.13	Female	Mcdonald Inc	Outlet	Christopherland	Michael Thomas	Christopherland
PROD39	STORE1	4	286.92	0.17	Credit Card	₹952.57	Female	Martinez, Adkins and Johnson	Flagship	Danielmouth	Jessica Gonzalez	Danielmouth
PROD88	STORE4	4	644.58	0.01	Credit Card	₹2,552.54	Female	Alexander Group	Online	Port Sharon	Wesley Short	Port Sharon
PROD34	STORE16	2	316.9	0.3	PayPal	₹443.66	Male	Frederick-Wright	Flagship	New Jimmymouth	Adam Castro	New Jimmymouth
PROD24	STORE15	4	403.47	0.29	PayPal	₹1,145.85	Male	Shelton PLC	Flagship	East Jenniferstad	Sherry Cabrera	East Jenniferstad
PROD36	STORE9	2	871.34	0.04	COD	₹2,509.46	Female	Moore, Padilla and Butler	Flagship	East Johnstad	Lindsay Beasley	East Johnstad
PROD65	STORE19	2	451.98	0.25	Credit Card	₹677.97	Male	Mann-Austin	Online	Vasquezborough	Arthur Warren	Vasquezborough
PROD27	STORE4	2	502.69	0.23	Credit Card	₹774.14	Female	Alexander Group	Online	Port Sharon	Molly Gardner	Port Sharon
PROD39	STORE4	2	559.23	0.25	COD	₹838.84	Female	Alexander Group	Online	Port Sharon	David Patterson	Port Sharon
PROD60	STORE13	4	513.98	0.04	COD	₹1,782.45	Female	Gardner, Humphrey and Contreras	Flagship	Johnsonville	Maria Hill	Johnsonville
PROD72	STORE17	3	986.04	0.27	COD	₹2,159.43	Female	Holloway Ltd	Flagship	Bensonhaven	Justin Cross	Bensonhaven
PROD43	STORE18	1	589.56	0.22	Credit Card	₹459.86	Female	Guerrero-Schmitt	Flagship	East Stevechester	Alexander Byrd	East Stevechester
PROD69	STORE10	3	979.42	0.03	PayPal	₹2,850.11	Female	Hill, Wagner and Ballard	Online	East Davidville	Elizabeth Foster	East Davidville
PROD12	STORE8	1	719.28	0.08	Credit Card	₹661.74	Female	Bean, Jones and Park	Online	Thompsonborough	Miranda King	Thompsonborough
PROD67	STORE6	3	77.6	0.07	Credit Card	₹216.50	Male	Benson-Elliott	Online	Port Josephberg	Jennifer Pruitt	Port Josephberg
PROD5	STORE3	4	852.71	0.14	PayPal	₹2,933.32	Male	Johnson-Wright	Outlet	Hoodberg	Erika Padilla	Hoodberg

Descriptive statistics value is found from total amount by calculating using formulas

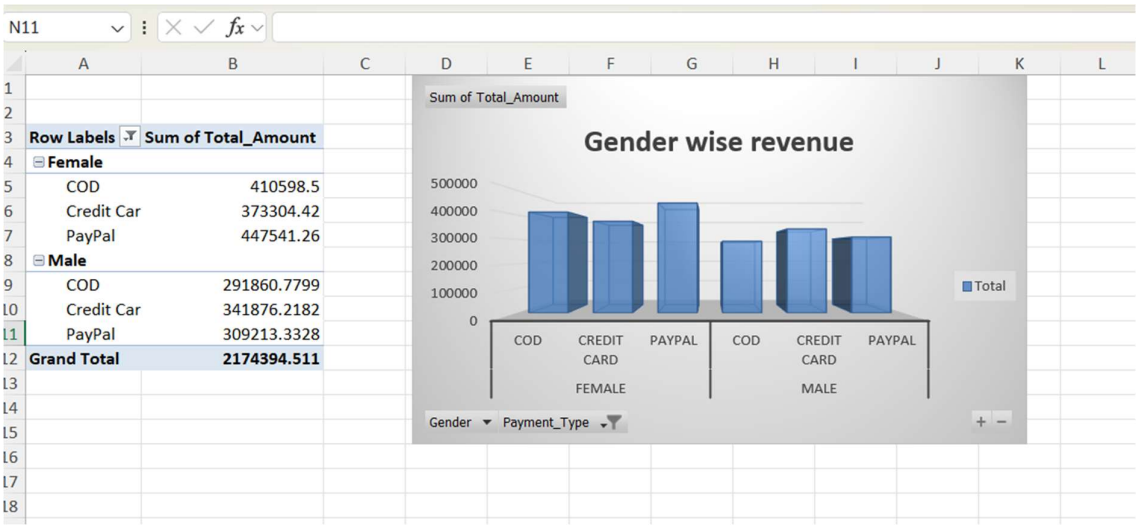


	A	B
4	Standard Deviation	819.41
5	MAX value	3908.24
6	MIN value	18.46
7	Median Value	868.98
8	Mode	502.11
9	Count	2001
10	Sum	2175481.7
11	Mean	1087.2

Store_Dim | Cleaned Store_Dim | Sales_Fact | Cleaned Sales_Fact- | Descriptive statistics

Pivot Tables and Charts

1) Gender wise revenue Pivot



Row Labels	Sum of Total_Amount
Female	
COD	410598.5
Credit Car	373304.42
PayPal	447541.26
Male	
COD	291860.7799
Credit Car	341876.2182
PayPal	309213.3328
Grand Total	**2174394.511**

2) Sales count for each product in year wise

Row Labels	Count of Unit_Price
2023	
Beauty	33
Clothing	37
Electronics	49
Home	49
Sports	49
2024	
Beauty	158
Clothing	181
Electronics	199
Home	203
Sports	258
2025	
Beauty	113
Clothing	116
Electronics	206
Home	158
Sports	191
Grand Total	2000



3) Amount generated based on store types in cities

Row Labels	Sum of Total_Amount
Flagship	
Bensonhaven	122176.5215
Danielmouth	101060.64
East Jenniferstad	99679.06
East Johnstad	110102.44
East Stevechester	104313.05
Johnsonville	114401.46
New Jimmymouth	111704.93
Washingtonfurt	85138.55
Online	
East Davidville	109791.7182
Lake Michael	93048.13
Port Josephberg	114533.75
Port Sharon	118814.11
Sabrinatown	116829.25
Thompsonborough	158971.44
Vasquezborough	94225.7028
Williamsfurt	108039.35
Outlet	
Christopherland	106248.24
Hoodberg	125011.81
New Michaelchester	81343.5584
North Nancyview	98960.8
Grand Total	2174394.511

